Quan (Kyle) Zhang

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Education

Ph.D., Economics (Quantitative Economics & Econometrics), UH Mānoa — Expected Dec 2027
M.S., Quantitative Health and Clinical Research, John A. Burns School of Medicine, UH Mānoa — Dec 2026
M.A., Economics, UH Mānoa — May 2026

Research Interests

Quantitative indicators of health and aging, such as frailty indices and other composite measures that combine many survey items and biomarkers into one score. Developing and validating new indicators where existing measures fall short, for other researchers to adopt and build on. Using these indicators to estimate causal effects as credibly as observational data allow. Methods include quasi-experimental designs, such as modern difference-in-differences and event studies; causal machine learning for heterogeneous treatment effects; causal mediation analysis of direct and indirect effects; and sensitivity analysis for unmeasured confounding. Applied econometrics and biostatistics for large national panels and health surveys. Related interests: global burden of disease; health policy and health equity; quantitative health sciences and clinical research methodology.

Peer-Reviewed Publications

1. WPR Antimicrobial Resistance Collaborators. The burden of bacterial antimicrobial resistance in the WHO Western Pacific Region 1990–2021: A cross-country systematic analysis with forecasts to 2050. *The Lancet Regional Health – Western Pacific*, 2026, 75, 101997. DOI
2. Yuan, K., Ye, Y., Zhou, X., Gu, Y., Jiang, Y., Yin, S., **Zhang, Q.**, Yang, H., Sang, X., Jiang, L., Zhuang, Z., Xu, K., & Huang, S. Integrated multi-omics and machine learning analyses to improve and explore molecular subtypes and immune profiles of primary hepatocellular carcinoma. *Discover Oncology*, 2026. DOI
3. Zhang, Q., & Fu, Y. Ensuring trustworthy AI assisted guideline development for clinical practice. *npj Digital Medicine*, 2026, 9(1), 661. DOI
4. Zhang, Q. Do negative social ties accelerate aging in adults, or does aging erode social ties? *Proceedings of the National Academy of Sciences (PNAS)*, 2026, 123(20), e2608036123. DOI
5. Zhang, Q. Keep REMS focused on safety, not as a barrier to generics. *Health Affairs*, 2026, 45(4), 451. DOI
6. Zhang, Q. Enhancing life-course analyses of financial disadvantage and depressive mood through longitudinal design and causal inference. *Journal of Epidemiology*, 2026, 36(5), 173–174. DOI
7. Zhang, Q. Overcoming bias in a fall prevention program in resource-limited areas. *JAMA*, 2026, 335(4), 370–371. DOI
8. Zhang, Q. Operationalizing distributional cost-effectiveness in genomic medicine: A United States–focused comment. *Value in Health*, 2026, 29(1), 172–173. DOI
9. Zhang, Q. On "The joint impact of greenspace and air pollution on mortality": Methodological proposals. *International Journal of Epidemiology*, 2026, 55(1), dyaf224. DOI
10. Zhang, Q. Re: Khodakarami et al, "Exploring neighborhood opportunity as a factor in pediatric asthma visits to the emergency department": A causal roadmap for Child Opportunity Index 2.0. *Public Health Reports*, 2025. DOI

Invited Book Chapters

1. Zhang, Q. "Social Determinants of Health and Multidimensional Poverty." In R. Baikady (Ed.), *The Palgrave Encyclopedia of Concepts and Theories in the Context of Global Poverty*. Palgrave Macmillan / Springer Nature, 2026. DOI
2. Zhang, Q. "Multidimensional Poverty Index (MPI): Measurement, Health Deprivations, and Policy Use." In R. Baikady (Ed.), *The Palgrave Encyclopedia of Global Poverty Research*. Palgrave Macmillan / Springer Nature. Under review.

GBD 2023 Collaborative Papers (Group Authorship, Under Review)

1. GBD 2023 Colon and Rectum Cancer Collaborators. "Global, Regional, and National Burden of Colon and Rectum Cancer from 1990 to 2023, with Forecasts to 2050: A Systematic Analysis for the Global Burden of Disease Study 2023." Under review, *The Lancet Oncology*. (IHME ID: 4219.)
2. GBD 2023 Type 1 Diabetes Collaborators. "Global, Regional, and National Type 1 Diabetes Burden: A Systematic Analysis of the Global Burden of Disease Study 2023." Under review, *The Lancet Primary Care*. (IHME ID: 5338.)
3. GBD 2023 Alzheimer's Collaborators. "Global, Regional, and National Burden of Alzheimer's Disease and Other Dementias in 204 Countries and Territories, 1990–2023, with Forecasts to 2050: A Systematic Analysis for the Global Burden of Disease Study 2023." Revised version under review, *The Lancet*. (IHME ID: 6217.)
4. GBD 2023 Diet Collaborators. "Global, Regional, and National Disease Burden Attributable to Dietary Risks among Adults 25 Years and Older, 1990–2023: A Systematic Analysis for the Global Burden of Disease Study 2023." Revised version under review, *The Lancet Public Health*. (IHME ID: 6115.)

5. GBD Non-communicable Diseases Collaborators. "Global Patterns, Trends, and Cross-Country Inequality in Non-Communicable Diseases and the Impact of Population Ageing: A Systematic Analysis for the Global Burden of Disease Study 2023." Revised version under review, *The Lancet Public Health*. (IHME ID: 5410.)

Working Papers

1. Zhang, Q. "Health Declines and Work at Older Ages: Evidence from China."
2. Zhang, Q., & Hu, W. "Subclinical Cardiometabolic–Inflammatory Burden, Disability-Free Working Life and Work Exit in Three Ageing Cohorts."
3. Zhang, Q., Lyu, J., Slavens, C., & Hu, W. "Sustained Insulin Resistance and Incident Coronary Heart Disease in Older English Adults: A Selection-Weighted Cohort Analysis of the English Longitudinal Study of Ageing." SSRN
4. Zhang, Q. "Health Shocks and the Permanent Labor-Market Exit of Older Europeans."
5. "Nutritional Depletion, Inflammation, and Colorectal Cancer in US Adults, 2001–2023."

Conference Presentations & Workshops

1. Zhang, Q. "Working Through Frailty: Chronic versus Acute Health Shocks and the Margins of Labor Supply." Southern Economic Association (SEA) 96th Annual Meeting, session "Health and Work," Houston, TX, November 21, 2026.
2. Zhang, Q. "Health Insurance Design and Work Exit After Chronic Disease Onset: Harmonized Evidence from the United States, England, and China." Gateway Research Workshop on Cross-Country Comparisons (Gateway to Global Aging Data), held during the Gerontological Society of America (GSA) 2026 Annual Scientific Meeting, National Harbor, MD, November 4, 2026.
3. AWARD Network Summer Training Institute (selected participant, one of 22 early-career scholars), University of California, San Francisco, San Francisco, CA, July 20–24, 2026.

Research Experience

GBD Senior Collaborator — Global Burden of Disease 2023, IHME, University of Washington 2025 – Present

- Peer-reviewed 13 GBD 2023 estimation manuscripts spanning cancer, cardiovascular and neurodegenerative disease, metabolic disease, mental disorders, dietary and environmental risks, antimicrobial resistance, and population ageing.
- Co-author, under the GBD collaborators' group authorship, of six GBD 2023 papers: one published in *The Lancet Regional Health – Western Pacific* (2026) and five under review at *The Lancet*, *The Lancet Public Health*, *The Lancet Oncology*, and *The Lancet Primary Care* (listed above).

Research Allied Health Professional — The Queen's Medical Center, Honolulu 2025 – 2026

- Built and audited an analytic colorectal-cancer cohort from the hospital cancer registry (2010–2022) for an IRB-approved study, and estimated disparities in early-onset diagnosis, stage, and survival with logistic and Cox models benchmarked against SEER; in collaboration with the Department of Quantitative Health Sciences, JABSOM.

Graduate Research Assistant — UHERO, University of Hawai'i at Mānoa 2024 – Present

Maui Wildfire Exposure Study (MauiWES) *2024 – Present*
NIH- and state-funded longitudinal cohort of 2,000+ participants, with UHERO, JABSOM, and UH Maui College.

- Supported data collection: data entry, on-site support at data-collection events, and descriptive analyses (distributions and summary statistics); acknowledged contributor to the team's 2025 *JAMA Network Open* study of health and social support after the Maui wildfires.

Kaiona Survey Project *2025 – 2026*
Youth health and well-being survey and program evaluation, in partnership with Wai'anae Coast Comprehensive Health Center.

- Built Qualtrics consent, demographic, survey, and intake instruments; cleaned and managed survey data; produced pre–post data reports for three youth summer programs.

Roots Reborn Program Evaluation *2024 – 2026*
Post-wildfire economic empowerment intervention.

- Led the evaluation of an IRB-approved program end to end: Qualtrics survey design, a linked before–after panel of 250+ participants, paired pre–post analysis, a cost–outcome framework, and an executive summary recommending the program's next steps.

Small Business Success in Honolulu Study *2024 – 2025*
City & County of Honolulu Office of Economic Revitalization. Co-author of the UHERO report *Investigating and Tackling Barriers and Needs for Small Business Success in Honolulu* (2025).

Teaching Experience

Instructor, ECON 321 Introduction to Statistics, Department of Economics, UH Mānoa Summer 2025
Teaching Assistant, Principles of Microeconomics (ECON 130) & Macroeconomics (ECON 131) 2023 – 2024

Awards & Honors

2026	Research Stipend, Department of Economics, University of Hawai'i at Mānoa
2026	AWARD Network Summer Training Institute, University of California, San Francisco (NIA/NIH-funded)
2025, 2026	Burnham Campbell Fellowship Award, Department of Economics, University of Hawai'i at Mānoa
2024 – 2026	Hung Family Fellowship, University of Hawai'i at Mānoa (AY 2024–25 and 2025–26)
2023	Achievement Award, Department of Economics, University of Hawai'i at Mānoa

Professional Service

Journal Peer Reviewer: *Public Health in Practice* (Elsevier), 2025–Present · *Aging*, 2026
Conference Abstract Reviewer: Gerontological Society of America (GSA) 2026 Annual Scientific Meeting, 2026 · Population Association of America (PAA) 2027 Applied Demography Conference, 2026
University Service: Reviewer, Graduate Student Organization Merit-Based Awards (Empowerment & Leadership category), University of Hawai'i at Mānoa, 2026

Professional Memberships

American Economic Association (AEA) · American Society of Health Economists (ASHEcon) · International Health Economics Association (iHEA) · Society for Epidemiologic Research (SER) · Gerontological Society of America (GSA) · Hawaii Pacific Gerontological Society (HPGS) · Hawaii Economic Association (HEA) · Pi Gamma Mu

Technical Skills

Statistical Methods: Generalized linear models (including logistic and penalized logistic regression), panel models with fixed effects, sampling methods for complex surveys, multiple imputation, mediation analysis, power and equivalence testing, etc. Highly proficient in data processing and analysis; quickly masters new statistical software and tools.
Causal Inference: Staggered difference-in-differences and event studies, instrumental variables, regression discontinuity, matching and inverse-probability weighting, g-computation, sensitivity analysis (E-values), etc.
Machine Learning: Causal forests, double/debiased machine learning, penalized regression (elastic net), cross-validation, etc.
Quantitative Health Sciences: Frailty index construction (deficit accumulation), survival analysis (Cox models), cost-effectiveness and value-of-information analysis, psychometric scale scoring, survey questionnaire design (Qualtrics), etc.
Programming and Tools: R (primary: data processing, modeling, visualization with ggplot2, reproducible workflows); working knowledge of SQL and Tableau; Git. With AI-assisted coding, quickly picks up other tools, such as Python, SAS, Stata, etc.
Data: National aging panels (HRS, ELSA, SHARE, CHARLS); NHANES, MEPS, IPUMS; SEER; hospital cancer-registry data, etc.

Research Training & Certifications

CITI Program: Responsible Conduct of Research; Biomedical Research Investigators and Key Personnel; Non-Exempt Biomedical Researchers and Key Personnel (incl. IPS); Good Clinical Practice (2025–2026) · IHME Introduction to GBD (2025)

Languages

English (fluent) · Mandarin Chinese (native)